

Consumer and Industrial IoT: A Systematic Review

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Abstract — The Internet of Things can be explained as the connection among millions of devices around us using various sensors and actuators, which results in endless applications such as industrial applications, smart city development, agriculture, automation, devices used for entertainment, and many others. Since IoT is a broad term and includes various applications, to avoid confusion divisions in IoT originated such as consumer IoT and industrial IoT. Some of the main applications in consumer IoT are smart home/city automation, health care devices and fitness trackers, personal asset tracking, VR Headsets, and many others. Earlier, International Data Corporation (IDC) in 2017 stated that consumer IoT / CIoT will be the highly used sector among IoT by 2020. Likewise, CIoT has been highly developed and benefitted by everyone around the globe. In a survey taken to understand the usage of CIoT, the smart home application is ranked first followed by wearables and smart city applications in the next two positions. This paper presents an overview of consumer IoT, various fields associated, and devices related to the consumer IoT. The comparison and trade-offs among devices are produced and also a short review on security issues in consumer IoT devices is brought out.

Keywords— *actuators, assets, CIoT, wearables, Internet of Things, comparisons.*

I. INTRODUCTION

Internet of Things (IoT) tethers various devices around the globe using the internet as a medium. With the help of the internet, the devices also exchange data among them [8]. In the 1980s the first IoT device was found by David Nichols. He invented a system that would report the contents of the coke machine to his fellow members. From there IoT tremendously developed year by year and has now covered all the sectors including ingestible sensors. It consumed an enormous effort to bring industrial IoT to existence. This is because, though IoT has various advantages there are few crucial points such as data storage, security, and analytics challenges. Likewise, different sectors in IoT have various advantages and fewer short comes which are to overcome in the nearing future.

The main reason for the origin of IoT devices is to reduce the human interference in operating devices around us. Home automation is a perfect example of this illustration. Instead of operating the appliances by people themselves, CIoT / consumer IoT provides automated applications which reduces time as well as gives notifications to people on every necessary operation it does [2]. As CIoT devices are user-friendly, people can configure the operations done by the device (for instance, switching ON light for a particular period of time). This

reduces energy wastage and enables the appliances to be used effectively.

The Internet of Things connects millions of devices around us. In order to connect these devices effectively, there are certain protocols that need to be adhered to. They are Wi-Fi, Bluetooth, Zigbee, MQTT, Near Field Communication (NFC), Z wave, Sigfox, and some others [1]. Among these, Wi-Fi, Bluetooth, and Zigbee are general-purpose protocols used in all kinds of wireless communication systems. MQTT is a message protocol that is also important in the smart city/home appliances. When designing devices protocols should be mainly considered because they pave the way for the devices to be flexible, user-friendly, scalable, and high quality with optimum power/energy consumption.

According to a survey by Statista in 2021 [4], In 2018, around 22 billion internet of things (IoT) devices were in use around the globe. As there are more people showing interest towards home automation and the need for smart appliances increases day by day, Forecasts implies that by 2030 around 50 billion IoT devices will be in use around the world. In an article [5], it is illustrated that IoT growth rates have been affected due to the pandemic situation in 2020. However, IDC expects the rates to return to normal in 2021 and also attain the Compound Annual Growth Rate of 11.3% during the period 2020 – 2024. According to an article published in June 2020 [6], industrial IoT ranks first among the applications of IoT followed by transportation, energy, smart city, healthcare, and some others. This survey taken by the article is based on the criteria excluding the consumer products such as wearables and smartphones. Whereas in [7], the growth of productivity and interest towards wearables is explained and said that by 2022, the wearable technology market is expected to be USD 51.60 Billion.

Despite the various benefits provided by IoT devices, there are few challenges in common to all IoT sectors. One such disadvantage is issues in interoperability [9]. Since IoT devices have their own hardware and software architectures, there may be a few restrictions among the devices to communicate. This can be solved by selecting a common standard for all operating devices in that particular network. By selecting this standard, it may also lead to the limitations in the functions performed by the system.

The hardware challenges pertaining to IoT are processing power, battery life, and memory/storage. The network challenges include scalability and bandwidth issues. Software challenges faced are fault tolerance in operating systems and dynamic security updates [10]. Among all these challenges the most important threat for IoT is its security. [10] has provided a few security issues and solutions for the same. The security issues concerning IoT applications such as home automation devices, health care devices are software bugs, web services, data, and cloud security. The solutions may include an end to end encryption, user authentication, and authorization. The issues in protocols such as ZigBee, sensors, and actuators are data sniffing and terminal security. The solutions are data encryption and providing sensor data security.

The foremost goal of this paper is outlined below:

- To provide a clear view on the basic concepts of IoT and its evolution.
- To list out the applications and various divisions in IoT.
- To explain some of the devices associated with health care and industrial applications.
- To provide a tradeoff/comparison among various fields in IoT.

II. RELATED WORKS

This element in the paper provides a description of the works associated with the applications in IoT.

Li et al. [11] discussed the problems involved in constructing reliable network computing signatures with numerous sources in the network and also proposed a method called multisource homomorphic network coding signature which can be used in decentralized IoT systems. Furthermore, an analysis on network coding signature, its construction, architecture was presented to understand the concepts more clearly. The forte of this study is to provide authentication of network coding in the standard model with a high-security scheme and to guarantee the availability in the multisource IoT system. The main disadvantage of this paper is that they didn't provide any experimental results besides mathematical proof.

Talavera et al. [12] directed an evaluation on the application of IoT in the fields of industrial and environmental agriculture. This paper studies the four main technological domains like monitoring (62%), control (25%), prediction (7%), and logistics (6%). It deals with two important subjects; the one is technological solutions that are mentioned and the other is about infrastructures. Based on the second study infrastructure and technology in industrial and environmental agriculture are grouped into sensing variables, storage strategies, power sources, communication technologies, edge computing technologies, actuators, and visualization strategies. For the mass advocacy in IoT, stronger standardization, scalability, security, power management, and the cost stand as barriers. The authors put forward the architectural diagram of the IoT-based industrial and environmental approach in the field of agriculture. The

proposed model includes four layers: application, physical, service, and communication. The backbone of this paper is the detailed explanation of the infrastructures, technologies, and architecture of the IoT context. The delicacy of the paper is that related works are not explained in detail. Khan et al. [13] presented an analysis on IoT security issues regarding architecture, protocols, communication, and networking in the field of blockchain and provided a solution for the challenges faced. The authors tabulate the issues from low level to high level with a solution. In addition to this, the background of blockchain, structural design of chained blocks, authentication, and privacy of the solution was explained. The implementation of the architecture is done with low power devices and cost. Although they fail to establish redundancy, the trade-off between reliability and cost for the whole infrastructure.

Hassija et al. [14] focused on the security-related challenges and threats in the applications of IoT. The authors discussed the high trust in emerging and existing technologies that can be implemented to resolve security-related threats. Blockchain, fog computing, edge computing, and machine learning are the technologies that are discussed in this paper. The visualization of present and future architectural diagrams is a remarkable one. It mentioned the sources of threat that are involved in the security threats: sensing layer, application layer, network layer, and middleware layer. Security issues in all these four layers were explained in a detailed manner. The supremacy of the review is proposing four different types of solutions; however, the affinity of each approach was not considered.

Chettri and Bera [15] reviewed low-powered WANs (wide-area network), its application in the outline of 5G IoT and challenges faced. Elaborately pointed out the seven main challenges and vision on 5G technology besides providing statistical data regarding IoT and its users in recent years. The architectural layers involved in 5G were network, application, sensor, communication, and architecture which are involved in piling up data, analyzing, refining, and exchanging them between gadgets and communication networks. Each layer and its respective models were elucidated in a precise manner. The main advantage of the paper is it discusses the research gap and future conduct. The main downside of the paper is that the authors failed to discuss the numerous hurdles in well-organized controlling and management of versatility in IoT networks.

Asghari et al. [16] give a detailed overview of the application of IoT in the fields like industry, smart city, healthcare, and the environment. Besides, the author provided the taxonomy of the IoT applications. The primacy of the review is that the author tabulated the review in a brief manner that is easy to read. The weakness of the work is that it fails to concentrate more on related works. Fig.1 depicts the divergent application of IoT casting-off in our day-to-day life, especially in the field of healthcare, industries, smart cities and home automation.

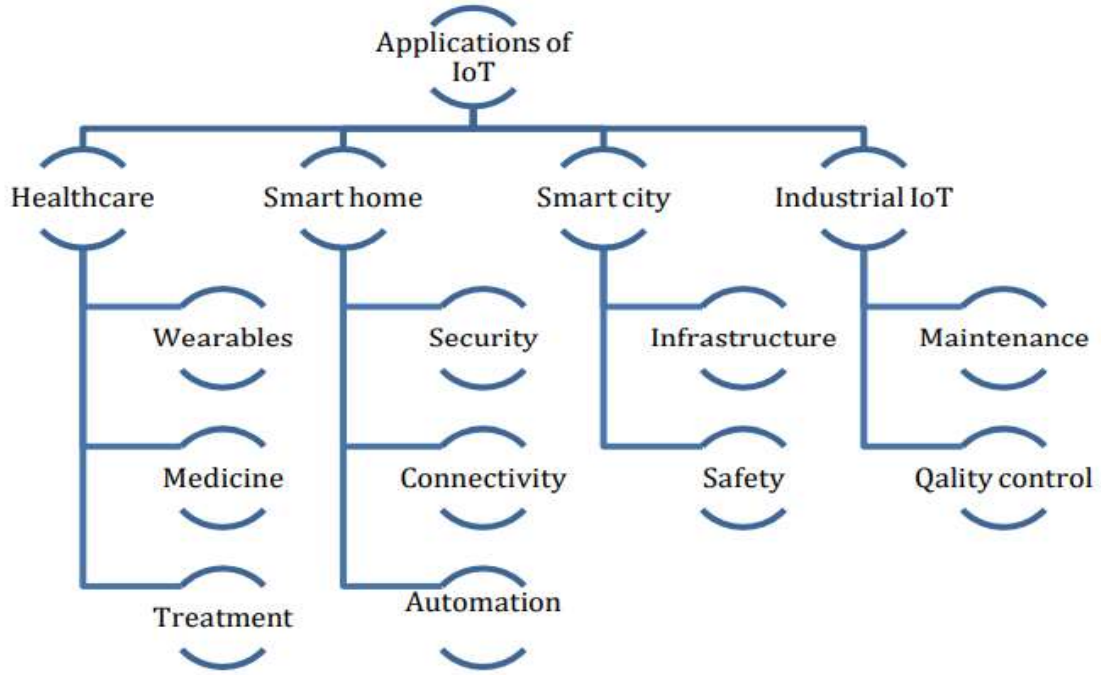


Fig. 1. Classification of applications of IoT

III. IOT APPLICATIONS

A. Smart city applications

Zia et al. [17] developed a constructive digital forensic methodology which instead of strengthening the safety and security of IoT, finds the provenance of attack and brings the offender in the due process with a proper digital affirmation. It approached four phases in the digital forensics: collection phase in which material data is spotted out, examination phase to find any piece of details which is relevant to the occurrence, analyzing phase which the most important one that helps to explicate the findings, and the final phase reporting where results acquired from the third phase were presented. The model discussed can be helpful in any kind of forensic analysis to gather and investigate the evidence.

Lingling et al. [18] developed an intelligent vehicle monitoring system that works on IoT. The architectural schematic of the proposed system contains five layers: application, network, data, object, and sensing layer. The system was first installed in the city of China for testing purposes. The technologies used here are GPS, RFID, video recognition to provide the information of the vehicle to the user.

Akbar et al. [19] come about with hail for estimating and contrasting diverse data flows for a dependable, effectual and expandable solution in the applications of IoT. In this manuscript, an architecture with two layers was put forward which has a layer to provide a synapse for a look into data from heterogeneous IoT structures in an expandable scheme, and the other layer for giving a probabilistic solution based on Bayesian network and CEP to relate high-level events. In Naranjo et al. [20] a novel architecture entitled Fog Computing Architecture Network (FOCAN) was developed.

This architectural schematic involves a multitier organization where the applications can be able to carry out on objects and communicate among them in the smart city conditions. The developed architecture decreases the delay and raises the productivity of services between objects with numerous provisions by refining energy provisioning. Alrashdi et al. [21] addressed the threats related to IoT cybersecurity in smart city applications and proposed solution anomaly detection which is based on machine learning algorithms. The smart city architecture proposed is based on fog computing to decrease latency between the cloud and the sensor. In [27] proposed an IoT testbed on all sides of the street lights using an architect. The advantage is that the model ensures end-to-end security.

B. Industrial applications

Abdullah et al. [22] developed an IoT code that schedules texts for appraising QoS recognition. In this work, texts are divided into two depending upon the urgency as high precedence and the best-effort messages. The main focus of this paper is to give a strengthened QoS-aware routing code in accordance to reduce power consumption. To decrease the delay of end-to-end broadcasting of information, the provided algorithm manages the fundamentals of IoT applications and initially, schedules abrupt demands with the highest supersede and then arranges the non-duty urgent messages concerning preventing famine in the dispersal of messages. MATLAB simulation was demonstrated to show the execution of the given retort.

Kiran and Rajalakshmi [23] proposed a novel Markov chain systematic model for unslotted CSMA/CA and PCA which can be used for industrial purposes. The above-mentioned model is additionally reached out to obtain the systematic model for slotted CSMA/CA and PCA. The advantage of the model is that it shows only 5% error during real-time testing. Huang et al. [24] provided a credit-based

proof-of-work model used in IoT devices that ensure the security of the system and transaction regulation concurrently. Wang et al. [26] designed a method based on edge computing to choose truthful and dependable service providers which solve the security and privacy-related issues. The proposed architecture consists of three parts namely, trust evaluation, record criterion, and service selection. Lightweight, implementation, and better security are the limitations of the system in edge computing methods.

Zhou et al. [25] depicted a structured way to acquire threshold by estimating the performance divergent computing manners that can be useful in real-time IIoT applications. The model provided is an efficient way to acquire threshold in the stochastic fluid model which can be further implemented in practical IIoT. Although it can be applied to small-scale applications.

C. Healthcare applications

Y. Lin et al [28] proposed a digital stethoscope that can be used to measure the ECG of a person and can envision the signals of the heart sound. ECG measuring device and heart sound measuring device are used along with a microphone and a circuit to record the sound and filter the external noises. The heart sound signal and ECG are transmitted to a person's smart device via an app and Bluetooth module. Using the app, a person can view the ECG signals and heart sound signals. The advantage of the model is the consumption of power is low and can be used to identify cardiovascular diseases.

In [29] the authors have developed an IoT monitoring system for health care. This model checks the biological factors of the patients using suitable sensors and transmits the data through the existing IoT usually available in homes. The proposed model is of low cost and alerts the doctor and the patient's relatives in case of any emergency. They used a smart chest strap for measuring the heart rate of the person. The presented prototype in [30] is used to monitor the health of the person by considering the predominant factors such as heartbeat and temperature. If there are any abrupt changes in the above-mentioned two factors the patient is alerted via smartphone using IoT technology. LCD, Wi-Fi, and Arduino Uno microcontroller are used in this model.

In [31] presents a secure healthcare monitoring system. The model uses a WMSN system and RFID body sensor because of its ability to allow smart solutions for many problems and the medical data of the person is wirelessly sent to the local workstation and the doctors can access the information whenever needed and the abnormalities in patient's condition are alerted to the doctor via SMS. The advantages of the proposed system are its security, reliability, and the patient's real-time monitoring. Ding et al [32] proposed a system of smart monitoring of health based on IoT technology. The model is an alarm-based monitoring system that alerts the doctor about the state of the person in case of

any emergency conditions. The information during the critical emergency is recorded in the hospital database. The advantage of the system is it is rapid and the power consumed by the devices in the system is low. The drawback of the system is the cost of the model is not taken into consideration.

S.M.R.Islam [33] described the broad overview of the contribution of IoT in the healthcare field. The authors have examined the recent development in IoT healthcare devices and have given a clear overview of their architecture, applications, and so on. And also, the paper evaluates the security threats in all those devices and the authors have come up with the model to reduce the risks related to privacy and security, etc. The author has proposed the algorithm to solve the security problems and also presented the challenges faced while implementing and developing the healthcare devices. The strength of this paper is the authors have depicted many images to make the readers understand the concept. The weakness of this work is that only an algorithm is given and the model is not implemented and verified.

D. Home automation

In [34] presented an IoT-based smart home system using two functionalities Philips's hue light bulb and Nest thermostat which can be used to do all the activities when we are away from home such as locking the door, switching on/off the light. This prevents home security-related issues. When the user gives the command to the JSON interface SHClient verifies the received information and executes the operation. There are many services such as alarm service, sensor service, lock service for changing the home smart. In [35] the author presented an organized methodology of smart homes based on IoT technology. Smart lighting, control of home appliances, home safety safeguard measures have been included in the design. The system functions have also been tested and with this system along with connectivity between several home appliances, the information between home and network can be shared by connecting the full system to the internet.

M.Al-Kuwari et al [36] presented a home automation design by sensing and monitoring some parameters like humidity, comfort lighting, temperature, etc based on IoT technology. For collecting the data and controlling the home appliances this design uses EmonCMS. A.Khan et al [37] proposed an IoT-based smart home design that uses a Wi-Fi-based microcontroller. Smartphones are used to control the smart home by turning on/off the AC depending on the temperature value. This design also gives remote control for those who can't use smartphones. The advantage of the design is it is very much useful for aged people and special persons. Table 1 gives a quick overview of the papers discussed. A brief description, advantages, and disadvantages of the systems reviewed have been listed below.

TABLE I. OVERVIEW OF THE STUDIES IN IOT APPLICATIONS

Reference no.	Brief description	Merits	Demerits
[17]	Digital forensic development	Helpful in investigations	IoT protocols
[18]	Intelligent vehicle monitoring system	Manage traffic	High cost
[19]	Bayesian network	Data management	High cost
[20]	FOCAN	Energy efficiency	Incompatible
[21]	Machine learning algorithm	Speed	Accuracy
[22]	Message sending algorithm	Performance	-
[23]	Markov chain	Less percentage of error	Low scalability
[24]	Blockchain transactions	Decreasing latency	Storage
[26]	Edge computing for security	Reduce workload	Implementation
[25]	Stochastic fluid model	Increasing service availability	High response time
[28]	Digital stethoscope	Low power consumption	Cost is not considered
[29]	Health monitoring system	Low cost	
[30]	Heart and temperature monitoring	Reduced time of sensing	Implementation
[31]	Secured health monitoring	Security, reliability	response time is not considered
[32]	Smart monitoring through smart devices	Rapid and low power consumption	Not considering cost
[33]	Overview of IoT healthcare devices	Simple and understandable	Only algorithm is mentioned
[34]	Smart Bulbs	Security	Cost is not considered
[35]	IoT based smart home	Storage	Implementation
[36]	Emon CMS based smart home	Simpler in design	Larger hardware
[37]	Home automation using Wi-Fi	Helpful for aged persons	-

IV. CONCLUSION

Since we are in the modern era there are an enormous number of applications/sectors in IoT. This paper tries to bring out the main sectors of IoT like smart city, health care, industrial, and home automation applications. In the course of this study, an extensive understanding of basic concepts of IoT is achieved, and also impact on the society by the major sectors in IoT is studied. By reviewing around 35 papers, this paper set forth 35% of smart city applications, 25% of healthcare devices, industrial applications, and 15% of home automation. Among the all reviewed, papers it is seen that huge importance is given to smart city applications, followed by equal share to healthcare and industrial IoT applications and some overview on home automation application. Form this study it is able to view the development of IoT in the past and future. Fig.2. displays pie chart of the content covered in this paper i.e., after analyzing the basic concepts of IoT and its applications. From the study it is clear that nearly 35% of paper deals with smart city applications followed by Health care and IIoT each 25% and Home automation constitutes the remaining 15%.

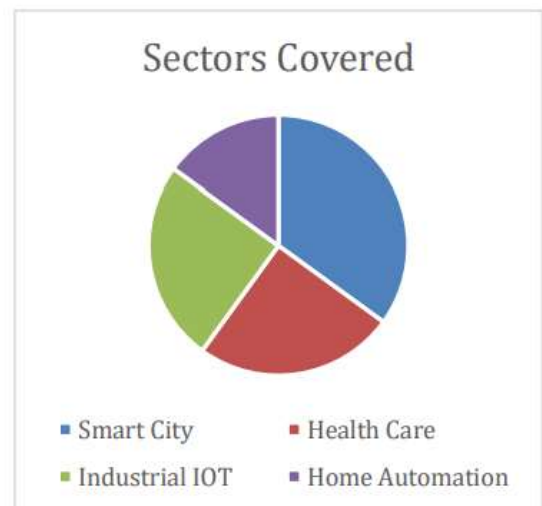


Fig.2. Sectors of IoT are covered in this paper.

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